Proactive Edge Attendance Tracker (PAT)

A high-performance facial recognition system using SCDFRN alignment and MobileFaceNet

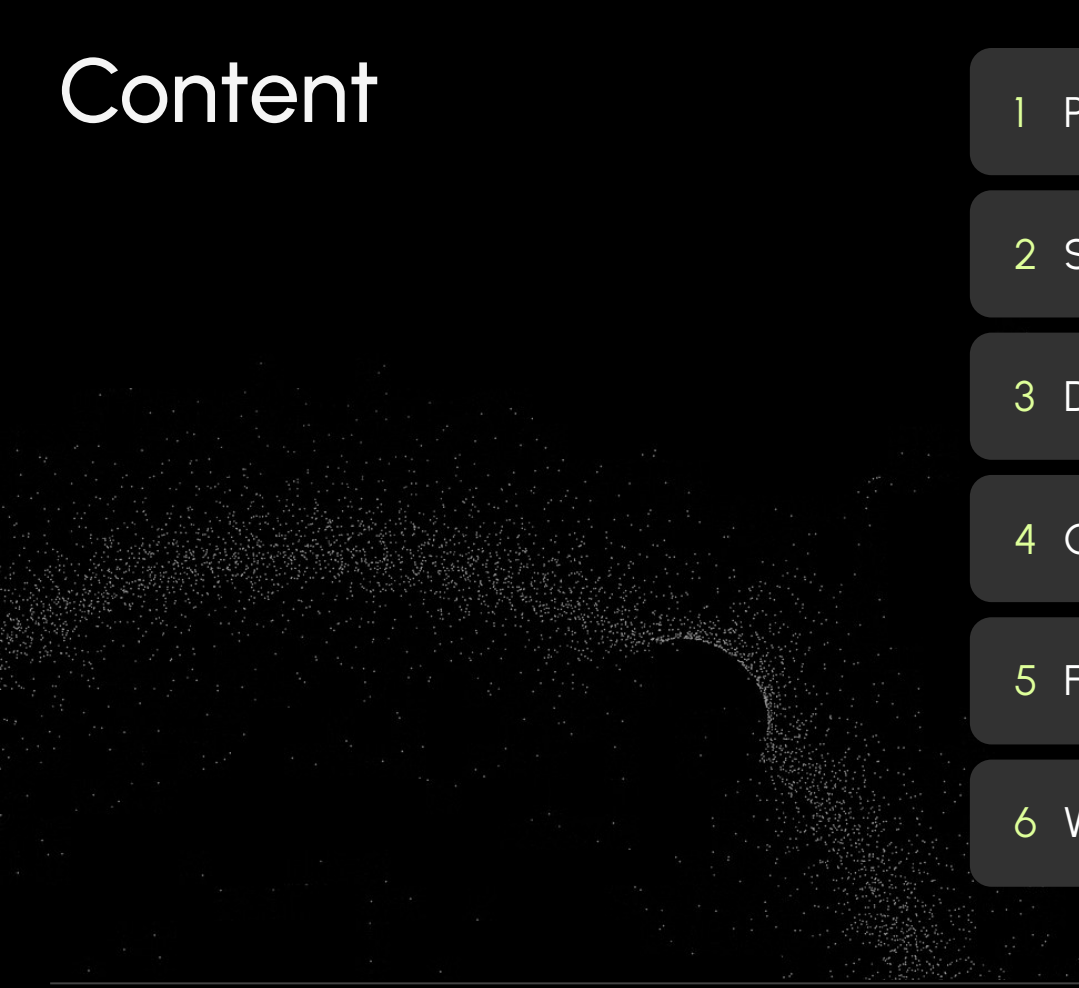
Winter 2025

Presented by Team 5

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Content



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2 Solution

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4 Challenges

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6 What we learned

There is an **Engagement Gap**

- **Widespread Absence ➜** Students frequently skip lectures due to low motivation or competing personal priorities.
- **Variable Instructional Quality ➜** Inconsistent teaching standards can inadvertently "justify" student disengagement.
- **Lost Opportunity ➜** Students miss the critical benefits of active learning and curated insights provided by thoughtful lecturers.



Traditional solutions

The **"Participation Grade"** Approach
To combat empty seats, lecturers often tether attendance to the final grade using:

- **Manual Tracking** ➜ Sign-in sheets and roll calls.
- **In-Class Deliverables** ➜ Pop quizzes and short written assignments
- **Digital Checks** ➜ Catch-all methods designed to force physical presence.



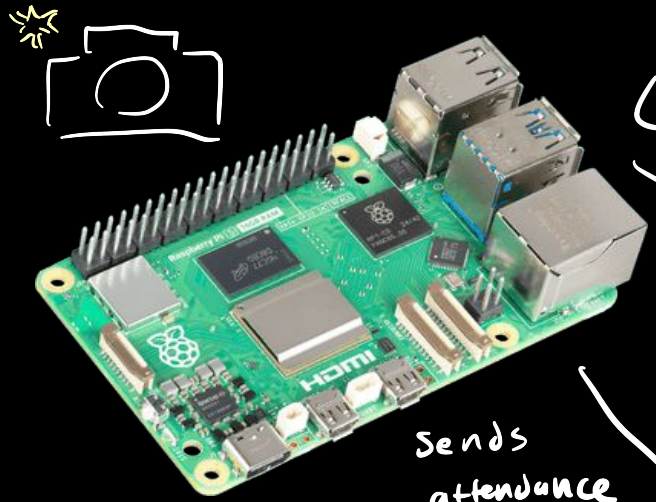
Tradition solutions are Ineffective

- **Gaming the System →** Students frequently "cheat" attendance via proxy signatures or sharing digital codes remotely.
- **"Ghost" Student →** Peers often leave immediately after signing in, skipping the actual learning window.
- **Conclusion →** The current model creates a culture of compliance rather than engagement, making it an ineffective solution for genuine learning.



Proactive Edge Attendance Tracker

- A distributed computing system that allows lecturers to bring an edge device into the lecture room to take automated attendance of course enrolled students through facial recognition.





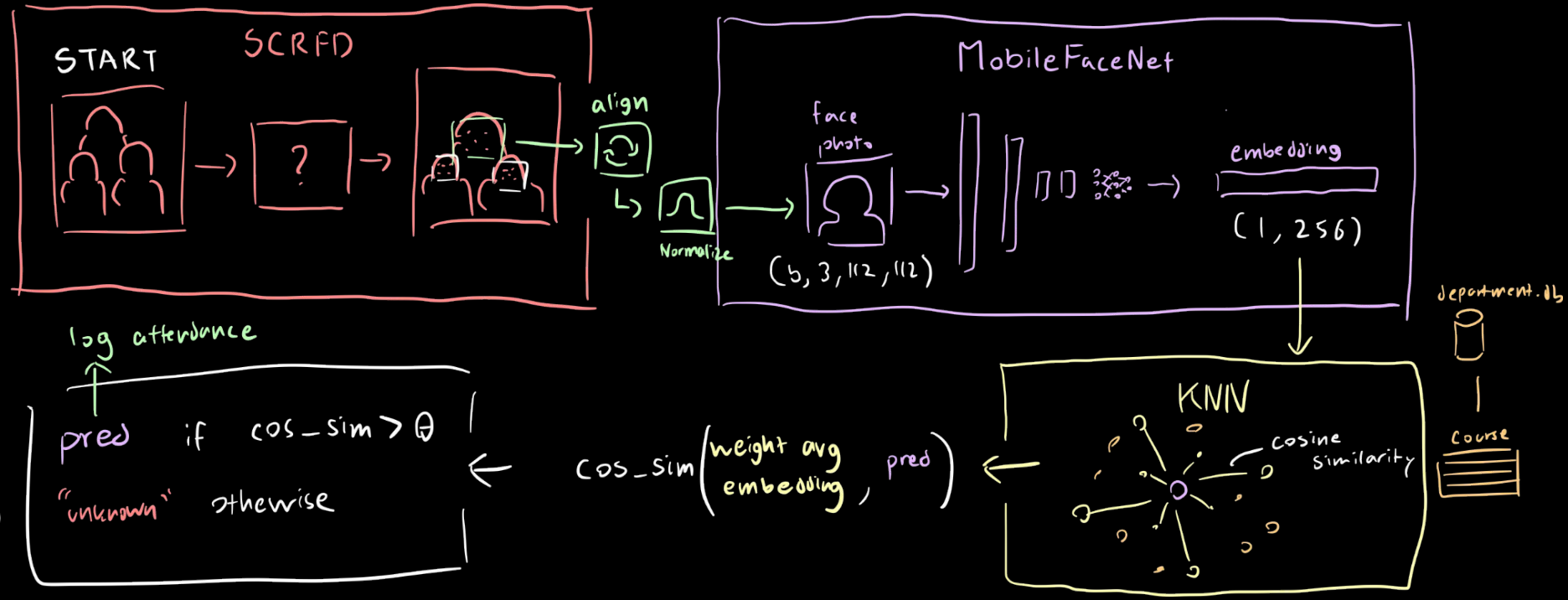


enrolled student embeddings

new student embeddings

sends attendance log

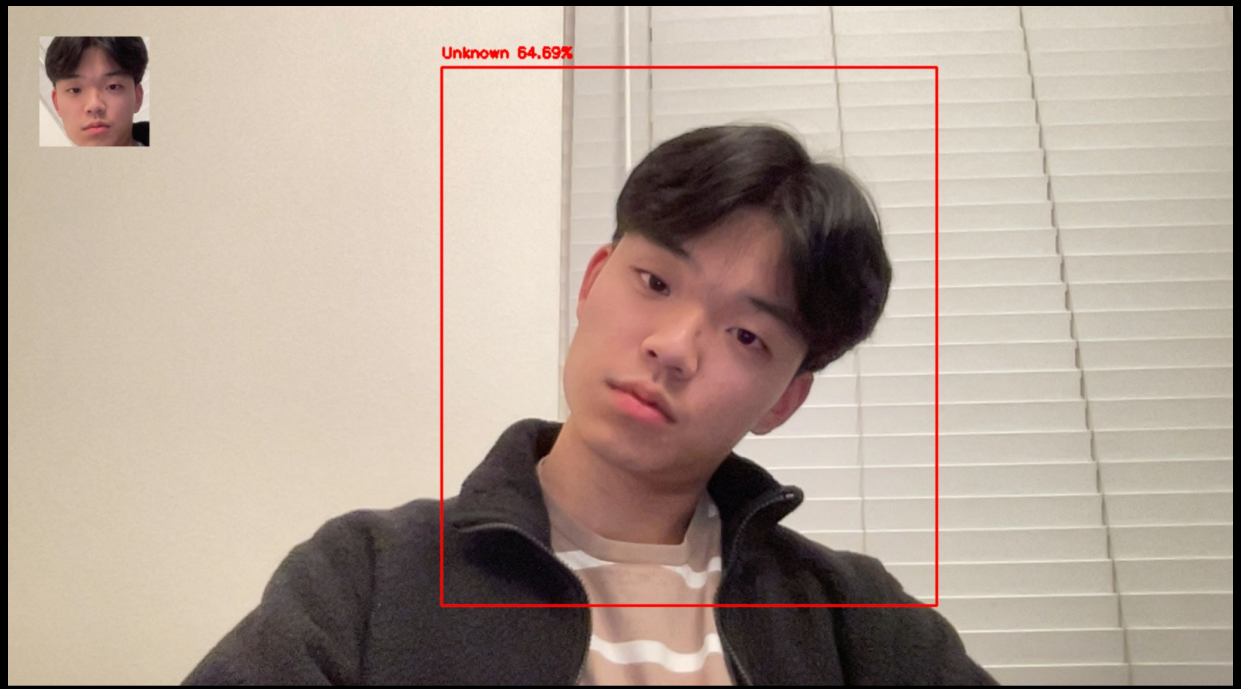
Facial Recognition Pipeline



Demo / Challenges



Challenges: Preprocessing



SEVERAL CAUSES RESULTING IN ALL FACES being cosine sim. of 0.9 or HIGHER:

- O.g. Bounding box captured only within the face
- alignment caused black edges (insufficient pixels)
- I had no clue that PyTorch face-net MTCNN did all that pre-processing.

SOLUTION:

- Increase bounding box
- Find Standard Face Landmark

Limitation / Future Improvements

- Implement easier method of capturing more/diverse photos of a single student
- Find estimated threshold using an ROC curve (extensive testing with face dataset) & evaluating performance of inference
- Offload and retrieve data to/from the cloud layer



Reflection on CS 131 Final Project

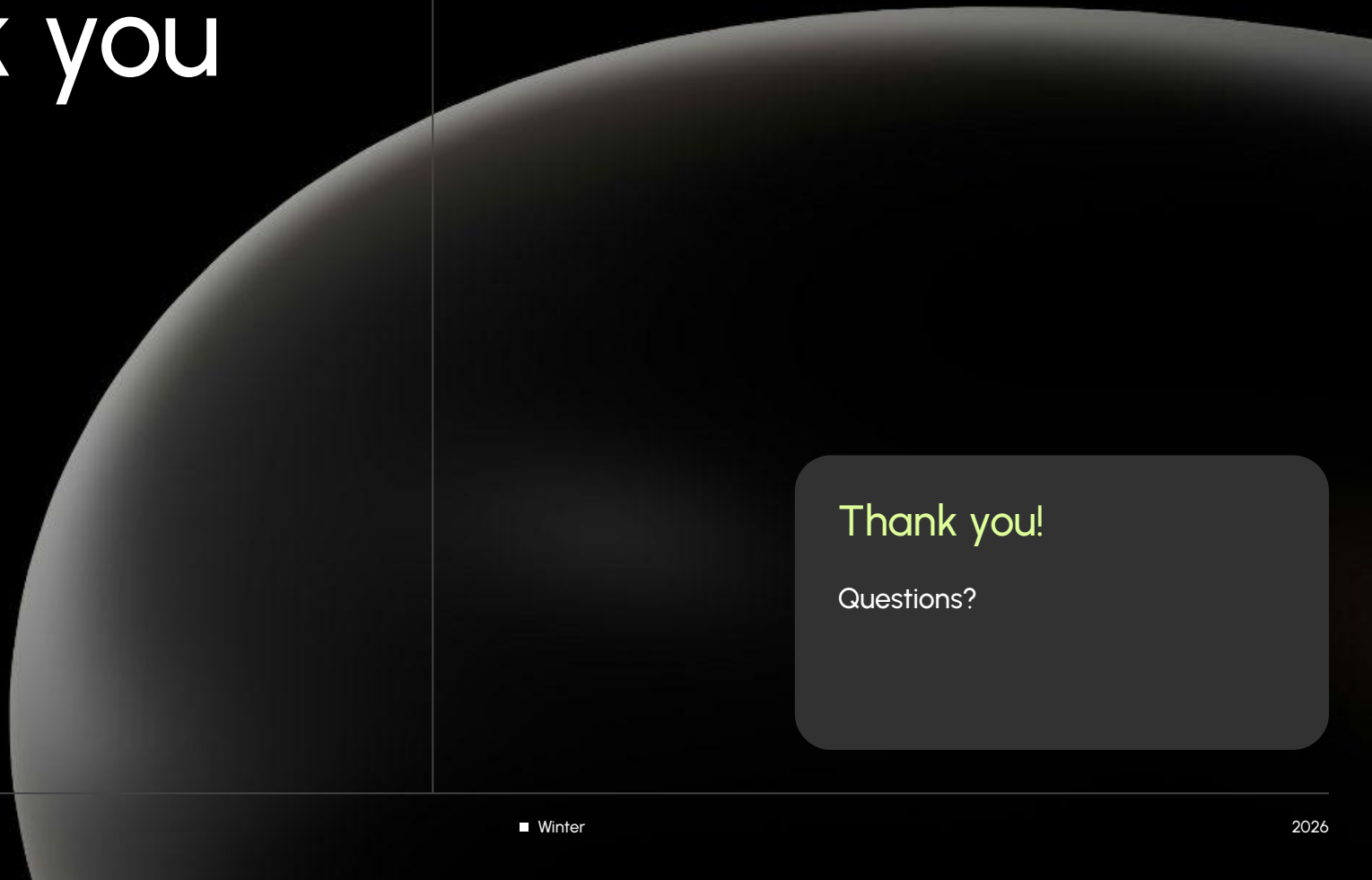
Jake Kim:

- Opencv-python & tensor manipulation
- Program Structure
- How to save, load, deploy models
- Docker Exposure
- Software Development is fun, might switch career paths

Stuart Arief:

- Python based project
- Hardware-Software integration
- Hands on experience with neural network models

Thank you



Thank you!

Questions?